INCH-POUND
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DETAIL SPECIFICATION

# TREADS, COMPOUND-FILLED, FOR INCLINED LADDERS

The specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers full treads and cap treads for inclined ladders that have slip-resistant compound-filled dovetail grooves in extruded aluminum bases.

1.2 Classification. Treads are of the following types, as specified (see 6.2):

a. Type I – Full tread with end plates.

b. Type IA – Full tread without end plates.

c. Type II – Cap tread.

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

| | | |
|---|---|---|
| MIL-STD-2151 | - | Inclined Ladder Tread Test Methods and Equipment for Wear, Slip-Resistance and Impact |

(Copies of this document are available online at http://quicksearch.dla.mil.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.2.2 Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9074-AR-GIB-010/278 - Requirements for Fabrication, Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

ASTM INTERNATIONAL

ASTM E648 - Standard Test Method for Critical Radiant Flux of Floor-Covering Systems Using a Radiant Heat Energy Source

ASTM E662 - Standard Test Method for Specific Optical Density of Smoke Generated by Solid Materials

(Copies of these documents are available online at www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO/IEC 17025 - General Requirements for the Competence of Testing and Calibration Laboratories

(Copies of this document are available online at www.iso.org.)

SAE INTERNATIONAL

SAE AMS4150 - Aluminum Alloy, Extrusions and Rings, 1.0Mg - 0.60Si - 0.28Cu - 0.20Cr - (6061-T6), Solution and Precipitation Heat Treated

(Copies of this document are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Material.

3.2.1 Bases. Extruded aluminum bases and end plates shall conform to Alloy 6061-T6 in accordance with SAE AMS4150.



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3.2.2 Filler. The filler material shall be an inorganic binder and abrasive grain composite formulated to comply with the physical appearance and performance as specified herein. Abrasive grain shall be 100 percent virgin grain aluminum oxide ($Al_2O_3$) with a Mohs scale hardness rating of 9. The grain quantity and size grading shall be selected to achieve the highest concentration of abrasive resistance. The filler shall be worked to provide a uniform, flattened surface that extends above the surface of the extrusion by a distance of 0.047 inch. The filler shall not overlap the groove edge by more than 0.031 inch. Exposed aluminum shall be cleaned of all extraneous filler material and the filler edges shall have a straight and uniform appearance.

3.3 Design and construction. Treads shall be of the design, construction, and physical dimensions as shown on figures 1 through 8. The type, width, and length shall be as specified (see 6.2).

3.3.1 Tolerances. Tolerances on dimensions as shown on figures 1 through 8 shall be as follows:

a. Overall length: ±0.063 inch.

b. Bolt hole diameters: ±0.016 inch.

c. Bolt hole spacing end plates: ±0.031 inch.

d. Groove width: ±0.016 inch.

e. Groove spacing: ±0.031inch.

f. Extrusion thickness: +0.010 inch.

g. Radii: ±0.010 inch.

h. Angles: ±1 degree, 0 minute.

i. Squareness of end plates: ±0 degree, 30 minutes.

j. Height of filler above extrusion: ±0.016 inch.

3.3.2 Color. The binding agent used in the filler shall be pigmented black. Aluminum shall remain self-colored.

3.3.3 Sealing. After fabrication, treads shall be given two coats of clear sealer to seal the filler and grooves against the intrusion of moisture. The sealer shall be silicon-based. When tested as specified in 4.5.4, treads shall not exhibit any signs of deterioration, cracking, or separation from the base extrusion.

3.3.4 Welding. Welding of end plates shall be in accordance with S9074-AR-GIB-010/278 and as shown on figure 1. Welding shall be from the underside of the tread and all excessive weld spatter shall be removed. End plates shall be smooth, flat, and square (see 3.3.1), and the edges shall be flush with the outside surfaces of the tread extrusion.

3.4 Performance.

3.4.1 Strength.

3.4.1.1 Type I and IA deflection. When tested in accordance with 4.5.1.1, deflection shall not exceed 0.10 inch and there shall be no permanent deformation.

3.4.1.2 Type I and IA local strength. After testing in accordance with 4.5.1.2, there shall be no signs of permanent deformation or visual damage.

3.4.2 Slip-resistant properties. When tested as specified in 4.5.2, the slip-resistance factor measured after 50,000 strokes shall be not less than the following values: 1.50 in the dry condition, 1.25 in the wet condition, and 1.00 in the oily condition. Maximum differential between the initial (before wear testing) and final slip-resistance factor for each condition shall not exceed 0.75.

3.4.3 Resistance to wear. Wear shall not exceed 0.010 inch after the first 10,000 strokes of the wear pads in accordance with MIL-STD-2151. Total additional wear, resulting from strokes of 20,001 through 50,000, shall not exceed 0.008 inch. Testing for wear-resistance shall be as specified in 4.5.3.



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3.4.4 Resistance to impact. When tested as specified in 4.5.5, the tread shall not exhibit deterioration or loss of the filler material deeper than 0.150 inch, cracks between impact points or projecting more than 0.50 inch away from the centers of impact, loosening or separation from the aluminum extrusion, nor shall any other area of the tread not previously subjected to impact testing show any signs of deterioration as a result of the test.

3.4.5 Fire performance. When tested as specified in 4.5.6, the tread specimen filler shall not exhibit destruction or deterioration, with the exception of change of color.

3.4.5.1 Critical radiant flux. The radiant flux shall be not less than 2.9 watts per square inch (0.45 watts per square centimeter).

3.4.5.2 Smoke density. The smoke density shall not exceed 450 (dimensionless).

3.5 Labeling. Each tread shall be stenciled on the underside with the following:

a. Manufacturer's name.

b. Manufacturer's address.

c. Year and month of manufacture.

d. National stock number (if applicable).

e. Type of tread and number of this specification.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on treads when a first article sample is required (see 3.1). This inspection shall include the examination of 4.4 and the tests of 4.5.

4.2.1 First article test specimens. The number of test specimens required for each test shall be as specified in table I. The tread selected for preparation of the fire test specimen (see 4.5.6) shall also be used for the examination of dimensions, shape, and tolerance of the extrusion (see 3.3 and 4.3.3).

TABLE I. First article specimen and test requirements.

| Performance characteristic | No. of specimens | Requirement paragraph | Test paragraph |
|---|---|---|---|
| Strength (type I and IA only) | 1 | 3.4.1 | 4.5.1 |
| Slip-resistance | | 3.4.2 | 4.5.2 |
| Wear-resistance | | 3.4.3 | 4.5.3 |
| Moisture and temperature resistance | 1 | 3.3.3 | 4.5.4 |
| Impact resistance | | 3.4.4 | 4.5.5 |
| Fire performance | As required by test method | 3.4.5 | 4.5.6 |

4.3 Conformance inspection. Conformance inspection shall be performed on the treads selected in accordance with 4.3.2 and 4.3.3. This inspection shall include the examination of 4.4 and the tests of 4.5.4, 4.5.5, and 4.5.6.

4.3.1 Inspection lot. Treads of the same type and width offered to the Government at one time shall be considered a lot for the purpose of inspection.



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4.3.2 Sampling for examination. Random samples of treads shall be selected from each lot offered to the Government in accordance with ANSI/ASQ Z1.4 at inspection level II and the Acceptance Quality Limit (AQL) specified (see 6.2).

4.3.3 Sampling for tests. One sample shall be tested in accordance with 4.5.4 and 4.5.5. A separate sample shall be tested in accordance with 4.5.6 and undergo the dimensional examination of 4.4.

4.4 Examination. Each sample selected, as specified in 4.2.1 and 4.3.2, shall be examined for the defects specified in table II. Failure to meet any requirement shall result in the rejection of the entire lot. Dimension and tolerance examination of the extrusion shall be conducted only on the specimen prepared for fire testing (see 4.3.3 and 4.5.6).

TABLE II. Classification of defects.

| Construction | Defects |
|---|---|
| Base | Material not as specified.<br>Type not as specified.<br>Length or width not as specified.<br>Dimensions or shape of extrusion (see 4.3.2) not as specified or within tolerance.<br>Mounting holes location and size not as specified.<br>Welding not as specified.<br>Outside surfaces not free of corrosion, scratches, roughness, blemishes, dirt or other extraneous material. |
| Filler | Material not as specified.<br>Cracks, loose material or inclusions of foreign material.<br>Color not as specified or nonuniform.<br>Sealant not as specified.<br>Height and overlap not as specified.<br>Nonuniform surface or edges. |
| Labeling | Missing or incomplete. |

4.5 Tests.

4.5.1 Strength tests.

4.5.1.1 Deflection test. A type I or type IA tread specimen of the longest length ordered shall be bolted to or supported by rigid end supports. A 300-pound concentrated load shall be applied to the center of the tread, distributed over an area 3½ inches wide by the width of the tread. After 10 minutes, the deflection at the center of the tread shall be measured. After removal of the load, the tread shall be inspected for permanent deformation.

4.5.1.2 Local load test. The type I or type IA tread specimen that was tested in 4.5.1.1 shall be tested by the application of a 200-pound load concentrated on a ½-inch square area. The load shall be applied at six different locations and held for 1 minute each. Each location shall be inspected for permanent deformation or visual damage after removal of the loads.

4.5.2 Slip-resistant properties (slip-resistance factor). The slip-resistance factor shall be determined for the tread sample selected as specified in 3.1 using methods and equipment in accordance with MIL-STD-2151. The initial slip-resistance factor shall be measured before the wear test specified in 4.5.3 and the final slip-resistance factor measured after 50,000 strokes. Measurements shall be made three times for each of the dry, wet, and oily conditions, and the average for each condition shall be recorded (see 3.4.2).



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4.5.3 Resistance to wear. Maximum average wear shall be determined for a tread sample using methods and equipment in accordance with MIL-STD-2151. Wear measurements shall be taken before testing and after the 10,000, 20,000, and 50,000 stroke intervals. Maximum average wear shall be determined for thee intervals using the methods in accordance with MIL-STD-2151.

4.5.4 Resistance to moisture and alternating temperature extremes. The sample treads shall be subjected to five cycles of immersion, cooling, and heating as follows:

a. Immerse the tread in room temperature tap water for a minimum period of 48 hours for the initial cycle. This period may be reduced to overnight for cycles two through five.

b. Drain the tread for 1 to 2 minutes, then place in 0±5 degrees Fahrenheit (°F) atmosphere for a minimum period of 4 hours ±5 minutes.

c. Remove and allow the tread to remain in room temperature atmosphere for 30 to 45 minutes, then place in 120±5 °F atmosphere for a minimum of 1 hour. Remove and allow the tread to return to room temperature before repeating step (a).

4.5.5 Resistance to impact. The tread sample that has been tested in accordance with 4.5.4 shall be mounted and tested for impact resistance in accordance with MIL-STD-2151. The ball shall be dropped from a height of 60 inches onto the tread in two separate patterns. The first pattern shall consist of nine impact points as shown on figure 9. The impacts shall be made in the order shown. The second pattern shall consist of three impact points centered on a filler strip and located near the longitudinal and transverse center of the top tread surface. The three points shall be separated each by 1 inch along the strip. Impact on the center point shall be made last.

4.5.6 Fire performance. Tread specimens in their end-use configuration shall be tested in accordance with 4.5.6.1 and 4.5.6.2.

4.5.6.1 Critical radiant flux. A specimen in its end-use configuration shall be tested in accordance with ASTM E648.

4.5.6.2 Smoke density. A specimen in its end-use configuration shall be tested in accordance with ASTM E662.

4.6 Fire testing provisions. All fire tests specified in this document shall be conducted by an independent testing laboratory that is accredited to ISO/IEC 17025 and is approved by the NAVSEA Technical Authority. Accreditation shall be obtained from a recognized accreditation body such as American Association for Laboratory Accreditation (A2LA) or International Code Council's International Accreditation Services (IAS). The scope of accreditation shall include specific flammability and fire tests. All other fire test provisions shall be as specified (see 6.2 and 6.5).

## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)



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6.1 Intended use. Compound-filled treads are intended for, but not limited to, all inclined ladders on board naval ships, including fixed and feathering tread accommodation ladders. Treads and ladders must be of high reliability commensurate with heavy usage and the service life of naval ships, and must be able to withstand and remain operable during and after shipboard fires, high sea states, and various impacts and loadings associated with naval operations. Current stock of compound-filled treads may be used until depleted; future requisitions for compound-filled treads should refer to the current version of this specification.

6.1.1 Type I and IA, full tread, figures 1 through 4. Full treads complete with end plates drilled with standard spaced holes, unless otherwise ordered, are used for mounting on new inclined ladders or for the replacement of old treads. They are suitable for both interior and exterior applications. The treads may also be used for fixed or feathering tread accommodation ladders. For feathering, treads must be ordered without end plates. Treads must not be installed in machinery spaces of U.S. Navy ships or other applications where the use of steel treads are required for resistance to fire damage.

6.1.2 Type II, cap tread, figures 5 and 8. Cap treads are used for mounting, without modification, over existing worn treads that are no more than ⅜ inch wider than nominal size. Cap treads may also be used for new installation when installed over steel channel sub-bases as in U.S. Navy ship machinery spaces or aluminum channel sub-bases. They are suitable for both interior and exterior application if adequate precautions are taken for the treatment of faying surfaces of dissimilar metals. Cap treads do not have end plates, but are drilled for mounting.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type of tread required (see 1.2 and 3.3).

c. Quantity required.

d. When first article inspection is required (see 3.1).

e. Length and width of tread required (see 3.3). Cap treads (type II) must be ordered 1 inch shorter than the treads being capped.

f. AQL of 1.0 (see 4.3.2).

g. Additional fire testing provisions (see 4.5 and 6.5).

h. Packaging requirements (see 5.1).

6.3 First article inspection. The Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Additional fire testing provisions. NAVSEA reserves the right to witness the tests, and perform any of the tests set forth herein where such testing is deemed necessary to assure compliance to prescribed requirements. NAVSEA will provide a letter upon request that indicates the approved laboratories which are accredited to ISO/IEC 17025 and which details how to obtain and maintain accreditation (see 6.2).

6.6 Subject term (key word) listing.

Fire-performance

Impact-resistant

Wear-resistant

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.



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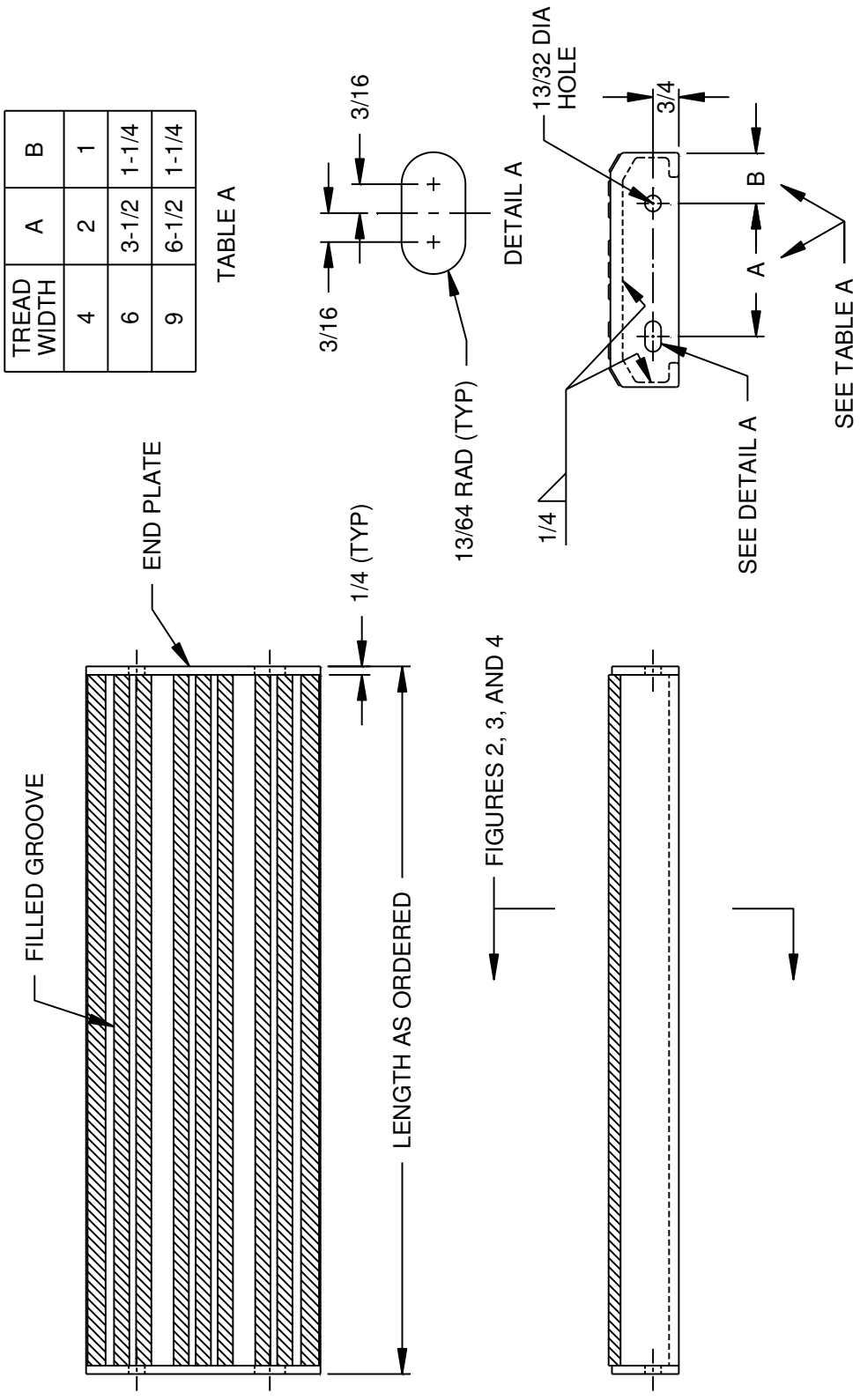
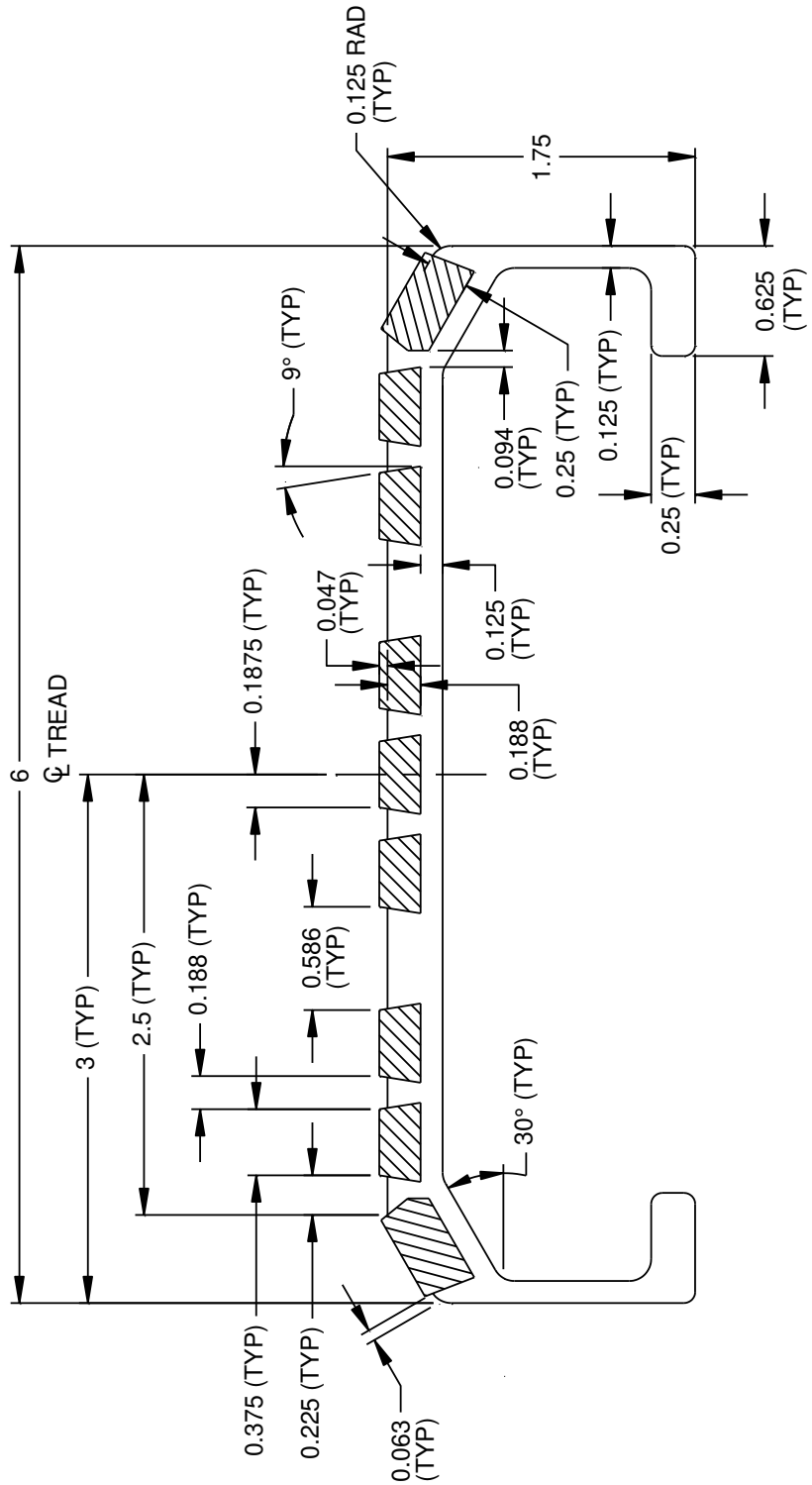



FIGURE 1. Type I and type IA full tread.



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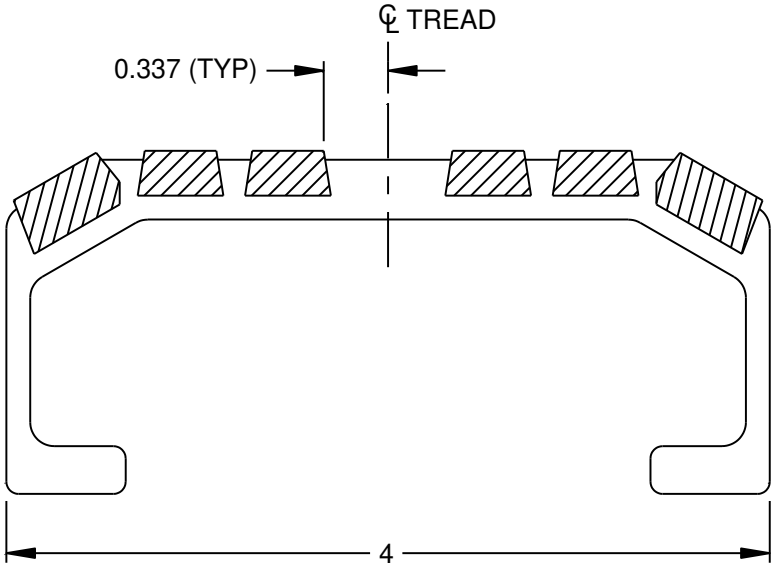

NOTES:

1. Fillets 0.125 RAD.
2. Rounds 0.063 RAD.

FIGURE 2. Type I and type IA full tread, 6-inch size.



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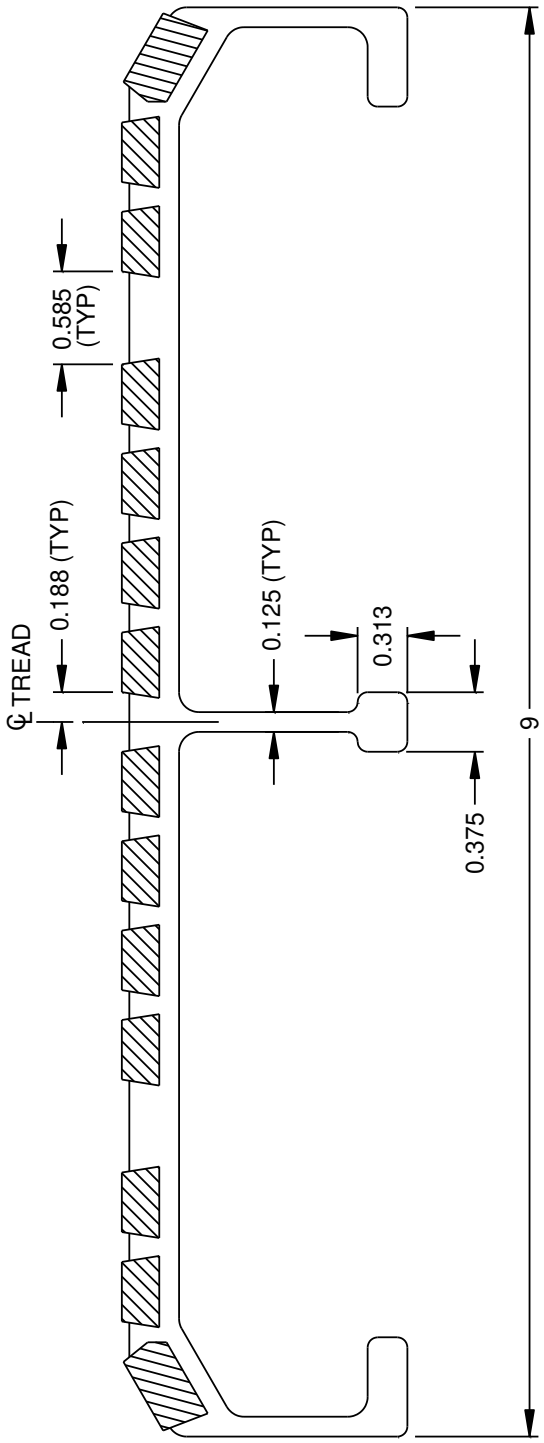

NOTE:

1. For information not shown, see figure 2.

FIGURE 3. Type I and type IA full tread, 4-inch size.



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NOTE:

1. For information not shown, see figure 2.

FIGURE 4. Type I and type IA full tread, 9-inch size.



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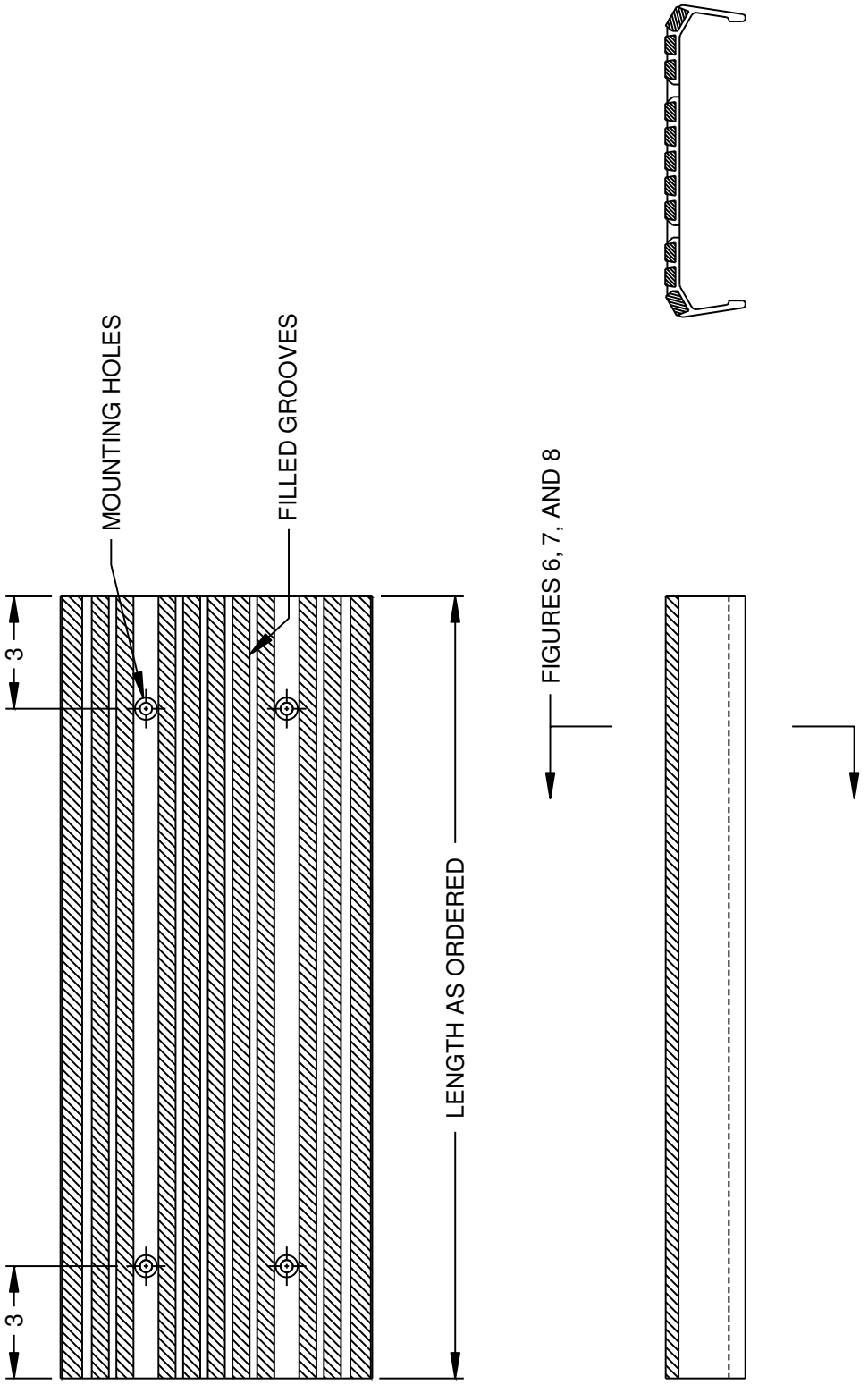
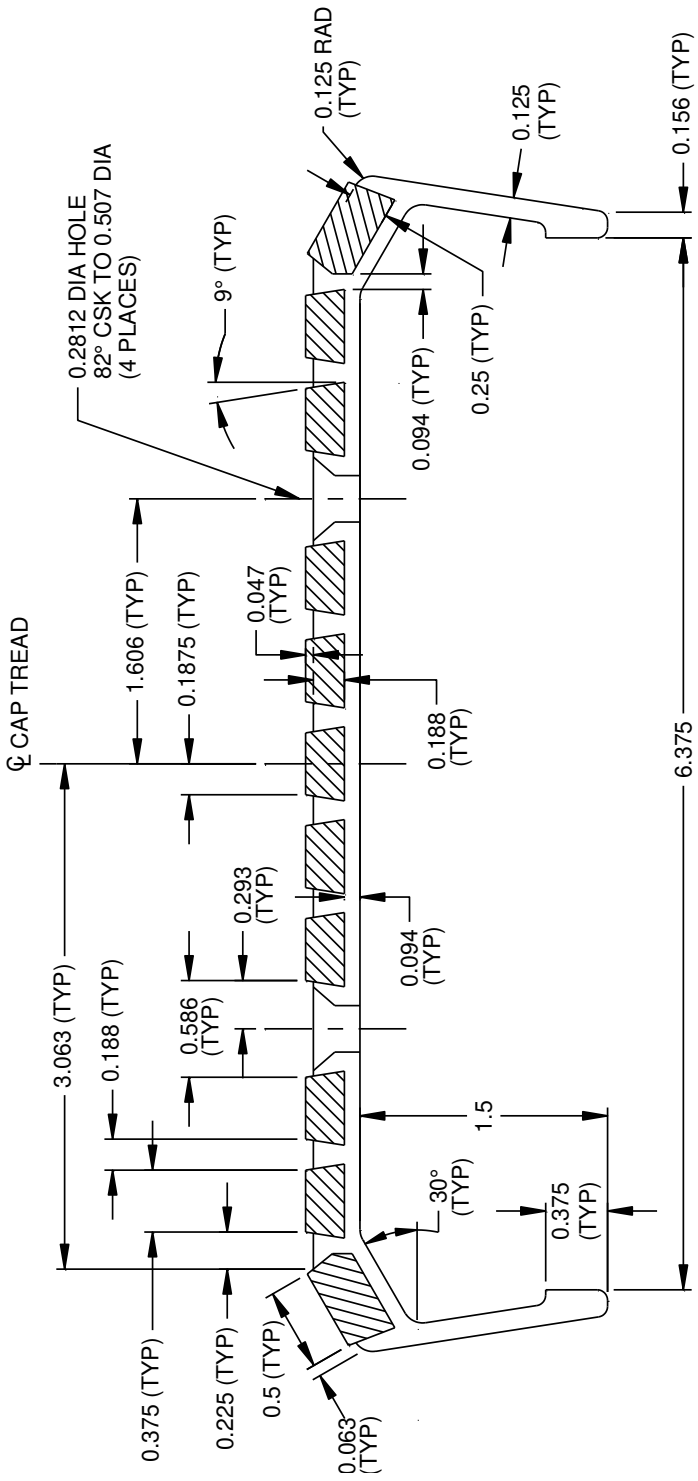



FIGURE 5. Type II cap tread.



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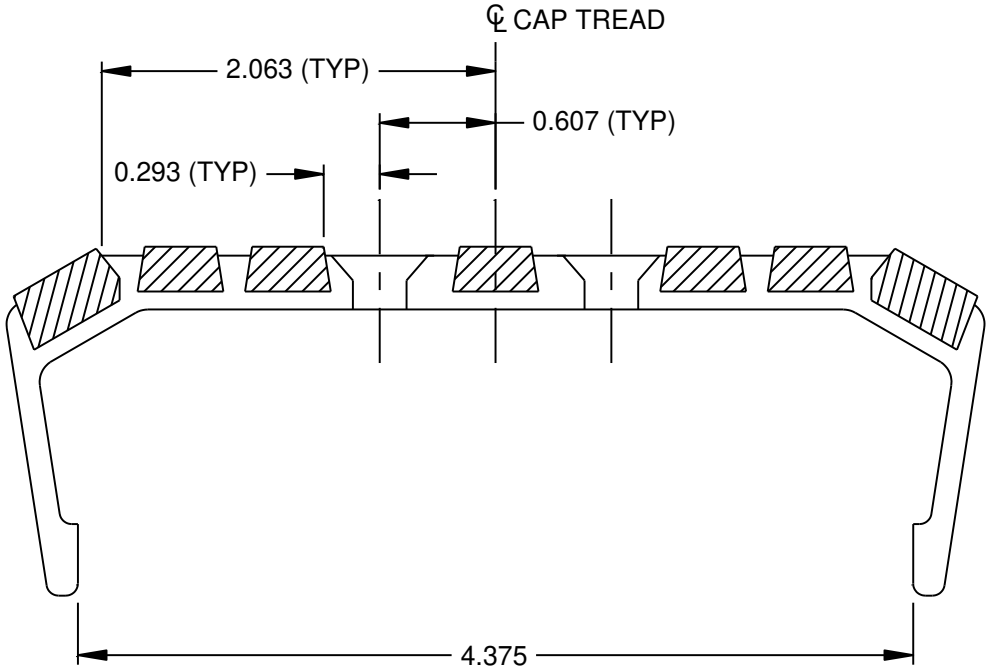

NOTES:

1. Fillets 0.125 RAD.
2. Rounds 0.063 RAD.

FIGURE 6. Type II cap tread, 6-inch size.



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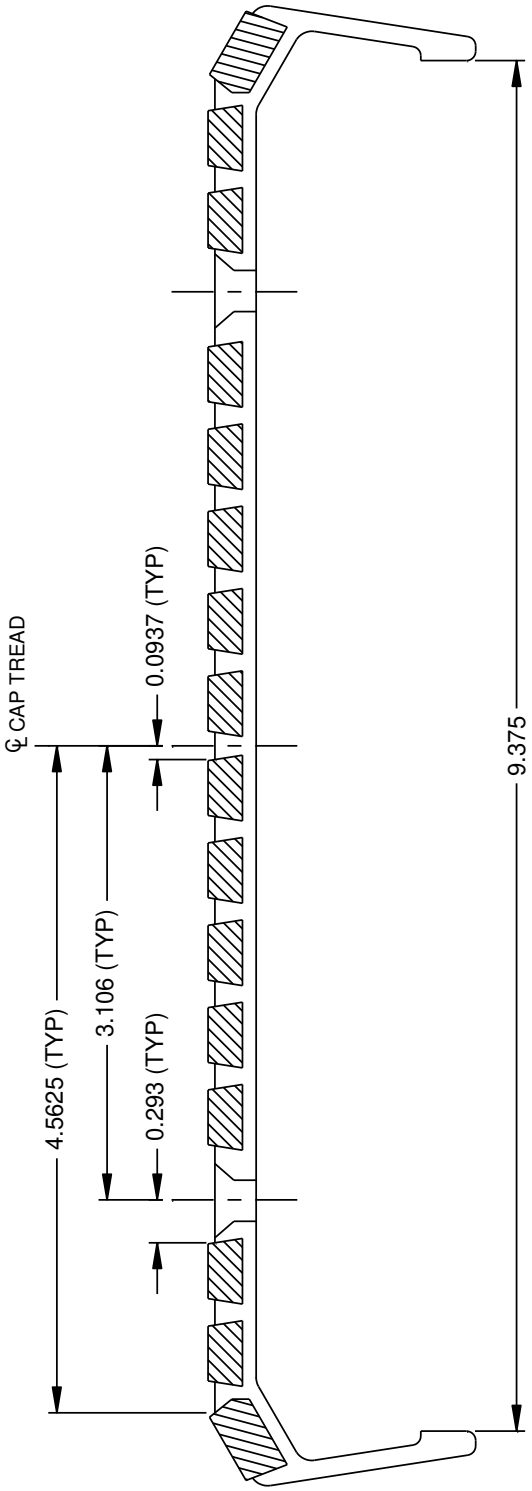

NOTE:

1. For information not shown, see figure 6.

FIGURE 7. Type II cap tread, 4-inch size.



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NOTE:

1. For information not shown, see figure 6.

FIGURE 8. Type II cap tread, 9-inch size.



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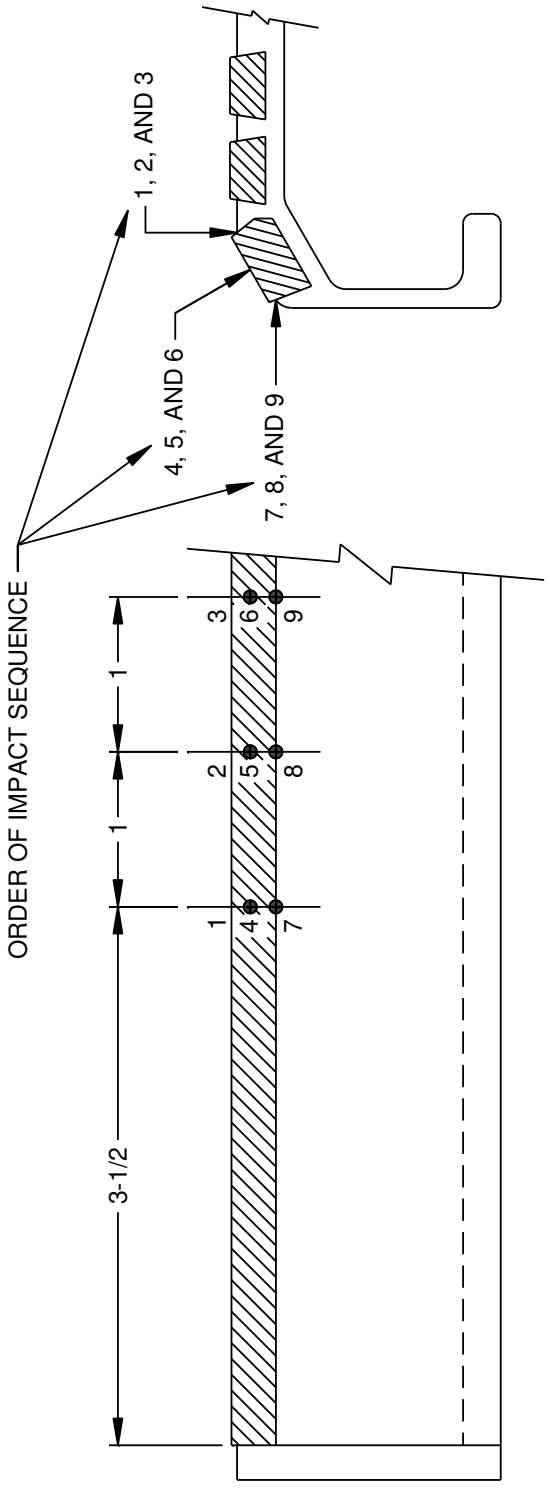



FIGURE 9. Impact test points.



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CONCLUDING MATERIAL

Preparing activity:
Navy – SH
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